

BÀI TẬP
Bài tập số 1
(Đề thi gồm có **01** trang)

Tên HP: **CƠ SỞ LẬP TRÌNH**

Mã HP: .. Số tín chỉ: ...

Học kỳ: **I** Năm học: **2024 –2025**

Ngày thi: **.../.../20...**

Thời gian làm bài: **100 phút** (không kể thời gian phát đề)

Câu 1 (2.0 điểm). Please read the following code snippet:

```
1  #include <iostream>
2  using namespace std;
3
4
5  int getPow(int x,int n) {
6      if (n == 0 && x == 0) return -1;
7      if (n == 0) return 1;
8      if (n == 1) return x;
9      int res = 1;
10     for (int i = 1 ; i <= n ; i++)
11         res = res * x;
12     return res;
13 }
14
15 int main() {
16
17     int n, k;
18     cin >> n >> k;
19     int *a = new int[n];
20     for (int i = 0; i < n; i++)
21         cin >> (*(a + i));
22
23     int j = n - 1;
24     for (int i = 0; i < j; i++) {
25         int tmp = a[i];
26         a[i] = a[j];
27         a[j] = tmp;
28         j--;
29     }
30
31     int s = 0;
32     for (int i = 0 ; i < n ; i++)
33         if (i % k == 0 || i == 0 || i == n) s += a[i];
34         else s -= a[i];
35     int res = getPow(s, k);
36
37     cout << res;
38     return 0;
39 }
```

Input từ bàn phím

6 2
2 5 4 3 4 5

- What is the output of the program?
- Please explain why the program produces this output.

Câu 2 (2.0 điểm).

Given an array **arr** consisting of n elements indexed from 0 to $n - 1$.

- Implement the function **int GcdOfArray(int *arr, int n)** which returns the greatest common divisor (GCD) of all the numbers in the array.
- A sequence of numbers is called "quasi-increasing" if there exists exactly one element that can be removed to make the array strictly increasing (a strictly increasing array satisfies the condition $arr_i < arr_{i+1}$ for all i). Implement the function **bool checkIncreasing(int *arr, int n)** to check whether the array **arr** is quasi-increasing.

Example: Given $n = 5$ and $arr = \{2, 4, 8, 6, 10\}$

- The function **int GcdOfArray(int *arr, int n)** returns the result 2.
- The function **bool checkIncreasing(int *arr, int n)** returns **true** because removing the number 6 transforms the array into 2, 4, 8, 10, which is a strictly increasing sequence.

Question 3 (2.0 points). Given a matrix **Arr** of integers with N rows and M columns, where rows are numbered from 0 to $N - 1$ and columns are numbered from 0 to $M - 1$. A submatrix is a rectangle within the original matrix, defined by a group of consecutive rows and a group of consecutive columns. In other words, a submatrix is represented as a rectangle with the top-left corner at (x_1, y_1) and the bottom-right corner at (x_2, y_2) ($x_1 \leq x_2$ and $y_1 \leq y_2$).

- Implement the function **void findColumnPrime(int **Arr, int n, int m)**. This function should output the indices of columns where all the elements in the column are prime numbers.
- Implement the function **int findMaxSumSubMatrix(int **Arr, int n, int m)**. The function's task is to find the submatrix with the maximum sum and return the sum of the numbers in that submatrix.

Example: Given $N = 4$ and $M = 5$, with the matrix **Arr**:

$$\begin{bmatrix} 2 & -4 & -5 & 2 & -5 \\ 7 & -7 & 3 & 2 & -4 \\ 3 & -6 & -4 & 5 & -3 \\ 5 & 5 & 3 & 7 & 1 \end{bmatrix}$$

- The output should be 0 and 3, which are the indices of the columns where all the elements are prime numbers. Specifically, column 0 contains (2, 7, 3, 5) and column 3 contains (2, 2, 5, 7).
- The answer is the submatrix with rows indexed from 1 to 3 and columns indexed from 0 to 3. This submatrix has a total sum of 23.

$$\begin{bmatrix} 7 & -7 & 3 & 2 \\ 3 & -6 & -4 & 5 \\ 5 & 5 & 3 & 7 \end{bmatrix}$$

Question 4 (2.0 points). Given a singly linked list, where each *node* contains an integer value x , you are provided with the head pointer of the list *head*.

- a. Write a function to create a new linked list containing only the *nodes* with values that are **prime numbers** from the original list. The *nodes* in the new list must maintain the original order of appearance.
 - *Nodes* that are not prime numbers will be removed.
- b. Write a function to **merge the consecutive nodes with prime number values into a single node**. The value of the new *node* will be the sum of the consecutive prime number values.
 - The new linked list will contain *nodes* with values equal to the sum of consecutive groups of prime numbers.

Input Data

The linked list is given in the form of a pointer *head*, pointing to the **first node** of the list.

Each *node* in the linked list has the following structure:

```
1 struct Node {  
2     int value;  
3     Node* next;  
4 };
```

Example

Input Linked List:

$2 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 10 \rightarrow 3 \rightarrow 9$

Part a: Create a new linked list containing only prime numbers.

$2 \rightarrow 5 \rightarrow 7 \rightarrow 3$

Part b: Merge the *nodes* containing consecutive prime numbers into a single *node*, where the value is the sum of those consecutive primes.

$2 \rightarrow 4 \rightarrow 12 \rightarrow 10 \rightarrow 3 \rightarrow 9$

Question 5 (2.0 points). Given a file 'INPUT.TXT' containing a list of information about items in a shopping cart on an e-commerce platform. Each line contains the following details: item code, item name, quantity, unit price, and total price. (The item code of different items may be the same.)

- a. Declare a structure data type with variables to store the following information: item code, item name, quantity, unit price, and total price. Then read the items from the 'INPUT.TXT' file into an array of this structure data type.
- b. Write a function to find the item with the highest total price. If there are multiple items with the same highest total price, select the item with the largest item code and save the result to the file 'OUTPUT.TXT'.
If there are multiple such items, print them all; if there are none, output -1 .

Example Illustration

INPUT.TXT	OUTPUT.TXT
32-Iphone 16-1-2300-2300 2023-Ipad pro M2-2-5000-10000 2023-Macbook pro M2-1-10000-10000 32-Iphone 5s-02-300-600	2023-Ipad pro M2-2-5000-10000 2023-Macbook pro M2-1-10000-10000

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Notes:

- Candidates are **not allowed** to use reference materials during the exam.
- Exam supervisors are not permitted to provide additional explanations.